

1-B:

Dynamic efficiency refers to the ability of a firm to reduce its costs **over time** through innovations and improvements in production processes arising from re-investment of super-normal profits in research and development. A downward shift in the average total cost curve represents dynamic efficiency, as the firm is able to produce the same quantity of output at a lower cost.

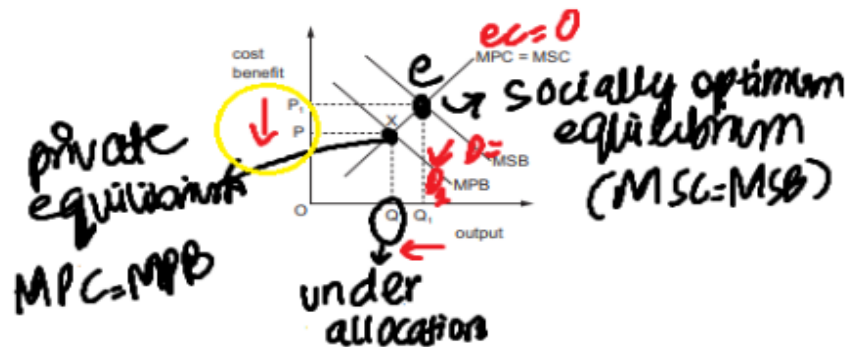
A. **Allocative efficiency** refers to the optimal allocation of resources in an industry, where resources are allocated in a way that maximizes social welfare.

C. **Pareto efficiency** is a concept that refers to a state where no individual can be made better off without making someone else worse off.

D. **Productive efficiency** refers to the ability of a firm to produce output at the lowest possible cost, given the current state of technology and resources.

Note: Productive efficiency is more focused on the firm's current production process, whereas dynamic efficiency looks at improvements over time. Also, productive and allocative efficiency are examples of **static efficiency**.

2-C:-



A, B: Since, external costs are shown to be zero, and the fact that Marginal Social Benefit is greater than Marginal private benefit, the case of “merit goods” is being presented.

And, **over-consumption, which results in over-pricing(key A), and over-production, which results in under – pricing(key B)** represents the case of “de-merit goods”.

As a result of which, options A and B, are incorrect.

C - : Since, external costs are shown to be zero, and the fact that Marginal Social Benefit is greater than Marginal private benefit, the case of “merit goods” is being presented.

And it can be seen in the diagram above, that **due to a lesser demand, these goods are under-consumed, and therefore under - priced**. This is why, key C, is correct.

D : Merit goods are also under-produced, and as a result of a fall in supply, they may be over-priced, but this is not seen in the diagram in question.

3 – D:

The table shows the results of a cost-benefit analysis of the construction of a new airport.

	US\$ million
private cost	400
private benefit	440
social cost	450
social benefit	480

What is the total value, in US\$ million, of all the externalities created by construction of the new airport?

A 30 B 40 C 50 **D 90**

$\rightarrow \text{social benefit} - \text{private benefit} \Rightarrow \text{external benefit}$
 $\hookrightarrow 480 - 440 \Rightarrow 40$
 $\text{externalities} = EB + EC \Rightarrow 50 + 40 \Rightarrow 90$

$\star \text{social cost} - \text{private cost} = \text{external cost}$
 $\hookrightarrow 450 - 400 = 50$

4- D:

Consumption combination "F" is not on the budget line but instead outside of the budget line, which means that it is un-attainable.

The same goes for consumption bundle "K", which is unattainable.

All the combinations inside of the budget line, and on the budget line are available/attainable combinations.

The points "G", "H", "L", and "J" (that represents the consumer-equilibrium where the indifference curve is tangential to the budget line, that is where, the utility-maximizing condition or the equi-marginal principle is established") are all "attainable combinations"

This is why, key D, becomes correct.

5 – D:

Consumer equilibrium is established when the equi-marginal principle is established, that is, when $MU_x / P_x = MU_y / P_y$

Rational consumer Romesh spent 90 rupees buying herb X and spice Y. Both cost 10 rupees per gram.

Multi-commodity equilibrium

MU_x / P_x	grams purchased	total utility herb X	MU_y	total utility spice Y	MU_y / P_y
1.5	2	15	12	12	1.2
1.3	3	30	13	25	1.3
1.5	4	45	10	35	1.0
1	5	55	7	42	0.7

$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$

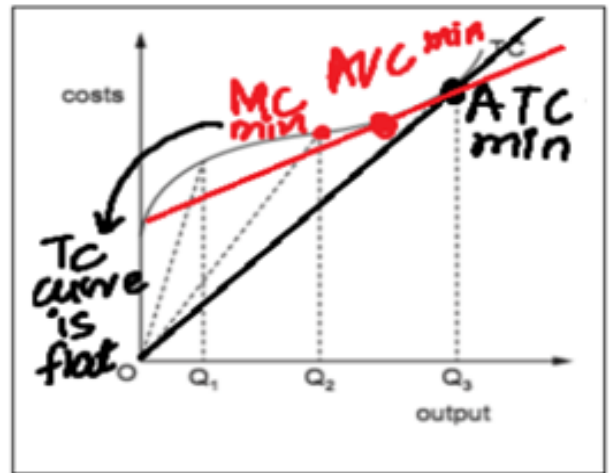
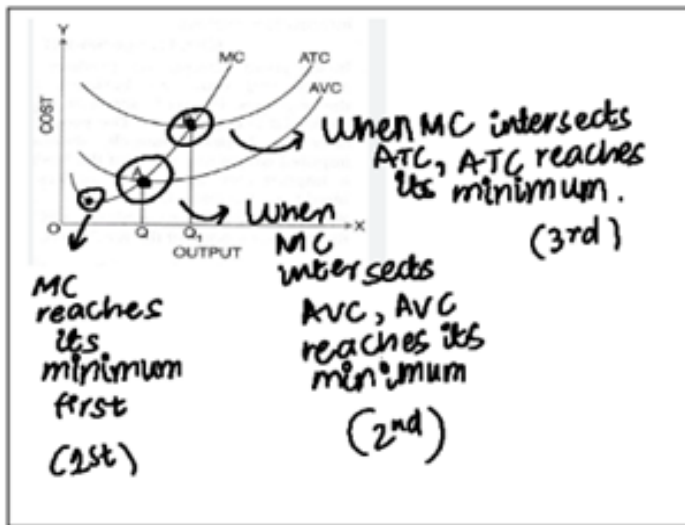
How much of each should he buy?

	herb X (grams)	spice Y (grams)
A	2	3
B	3	5
C	4	5
D	5	4

$\star$ equi-marginal principle is established when:

$\star$ 5 herb X & 4 herb Y consumed

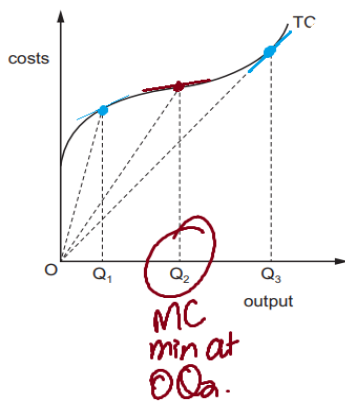
6 – A:



A – Average total cost is minimized at output “OQ3”. This is because the line passing from the origin is tangent to the total cost line.

B , C – Average variable cost is minimized in between OQ2 and OQ3. This is because the line below the total cost curve is tangent to the total cost curve.

D – Marginal cost is minimized at output OQ2(see below)



*To check the level of output where Marginal cost is the lowest, we need to see where the slope of the total cost curve is the flattest

* For this, a line will need to be drawn which is tangential to the total cost curve [corresponding to each output level]

* It can be seen that the slope of the line [corresponding to the output level OQ2] is the flattest, which represents that the marginal cost [the change in total cost] is the lowest.

7 – C:

C - Cartels are agreements between companies to fix prices, restrict output, or engage in other anticompetitive behaviors. They are illegal in many countries and are often subject to investigation and penalties by government agencies responsible for enforcing antitrust or competition laws.

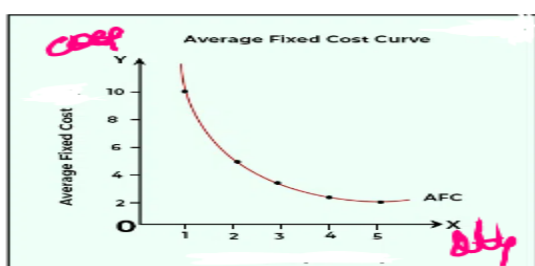
A) Cartels are not a form of trade union restrictive practice. Trade unions represent workers, whereas cartels are agreements between companies.

B) Cartels are not a type of industrial merger. A merger is when two or more companies combine to form a new entity, whereas a cartel is an agreement between separate companies to coordinate their behavior.

D) Cartels do not establish a vertical link between firms. Instead, it establishes a horizontal link between firms.

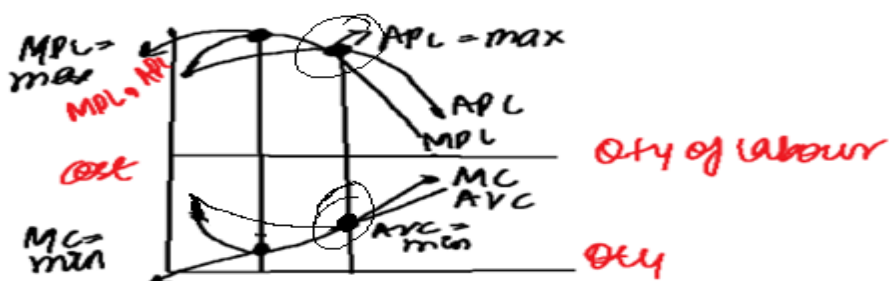
8- C:

A: This is incorrect as the average fixed cost always keeps on falling as output increases, which is why, the Average fixed cost curve is downward sloping.

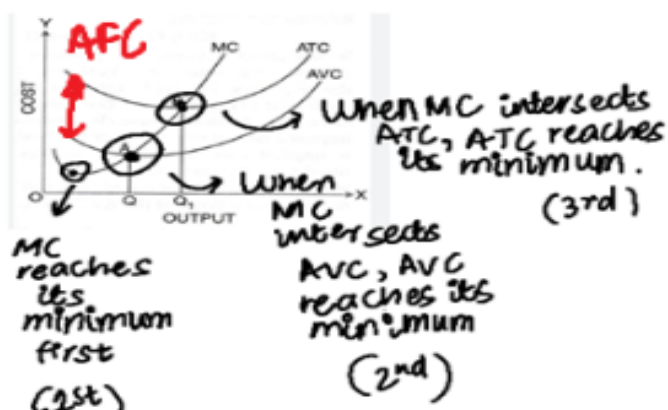


B: It can be seen in the diagram below that the average variable cost increases when the average product of labor falls, and when the marginal cost is greater than the average variable cost.

Note: The law of diminishing returns begins to operate when the marginal product of labor falls. This explains why key B is incorrect.



C: It can be seen in diagram#2 that the minimum point of the average total cost occurs at a greater volume of output than the minimum point of average variable cost. This is why, key C, becomes the answer.

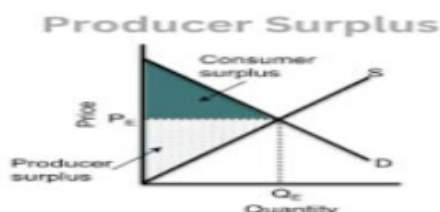


Diagram# 2

D: It can be seen in the diagram above that at the point where the marginal cost reaches its minimum point, Average fixed cost that is the vertical distance between the ATC curve and AVC curve is very high, and is therefore greater than Average variable cost.

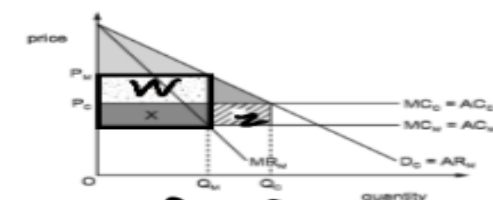
9- B:

Producer surplus is the difference between the minimum price a producer is willing to accept, and the price that he ends up charging. It is the area above the supply curve, and below the price line "P" This can be seen in the diagram below:



→ Here, in this case of imperfect competition, we will have to "assume that Marginal cost = Supply", in order to determine the producer surplus.

This means that we are interested in an area below the price line "PM" and above the marginal cost curve "MCm", at output of "Qm"



* below P_m
 * Above MC_m
 * At Q_m output

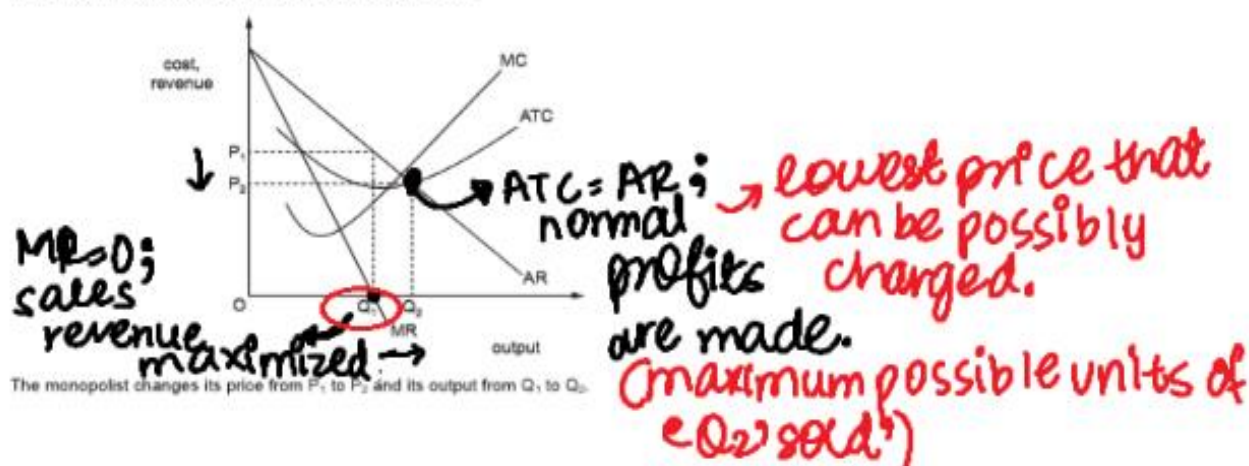
Therefore, the producer surplus is "W+X", which means that, option B is correct.

10- B:

- A – When large firms charge higher prices, they are responsible for a reduction in consumer surplus as they convert consumer surplus into producer revenue. And, as a result of high producer revenues, they benefit because of a likelihood of a rise in “profits”
- B- If the cost of funding the merger is very high, then the merger may fail as it would lead to firms experiencing cash flow problems and in turn liquidity issues. This is the correct option.
- C- If because of merger, the large firm is able to minimize its Long run Average cost and in turn benefit from economies of scale, this represents the success of the merger.
- D – Because of market dominance, when large firms charge higher prices, they are responsible for a reduction in consumer surplus as they convert consumer surplus into producer revenue. And, as a result of high producer revenues, they benefit because of a likelihood of a rise in “profits”

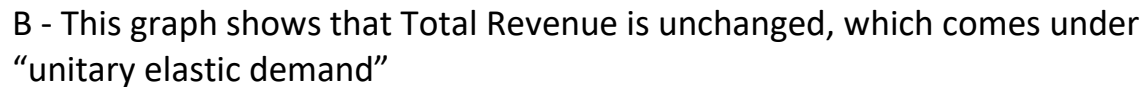
11- D:

The diagram shows a monopolist's cost and revenue curves.



Here, at Q_1 , since $MR=0$, this means that “sales revenue maximization” is achieved. And, at Q_2 , since $ATC=AR$, this means that “normal profits” will be made, and since here, the lowest possible price is charged, this means that the output or “the units sold will be maximized”. This is why, no other option, except for D can be correct.

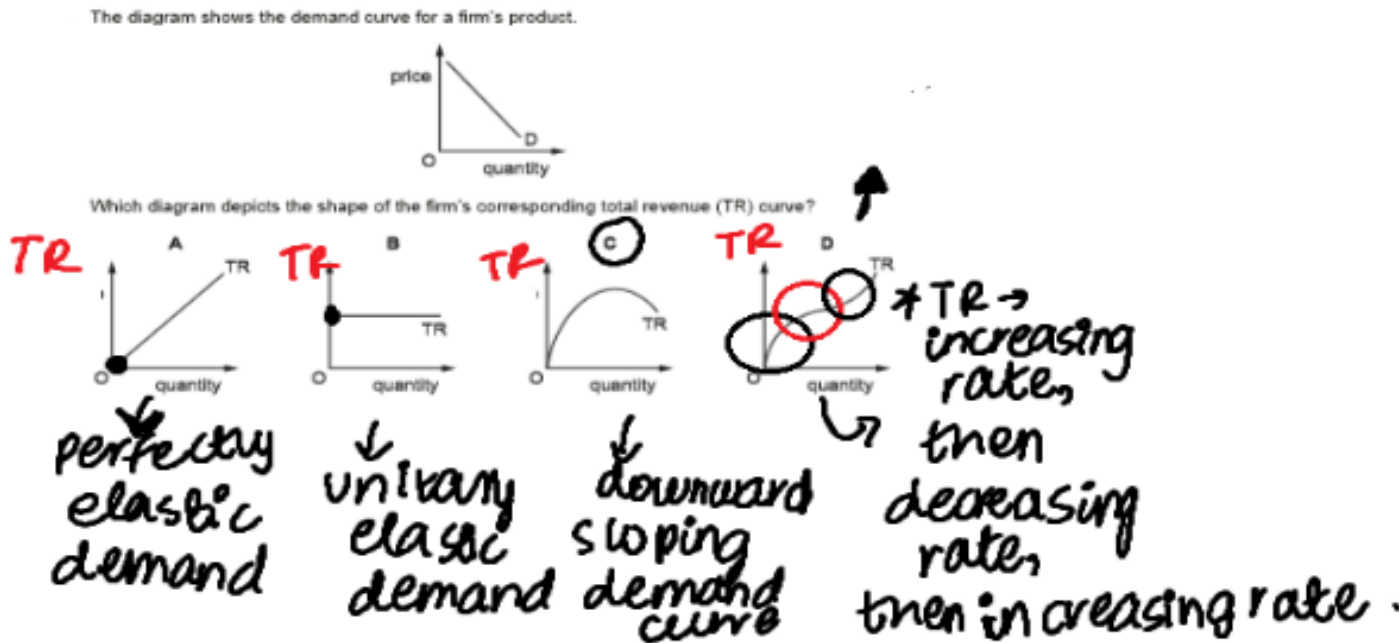
A – Key A represents the total revenue graph of a perfectly competitive firm.



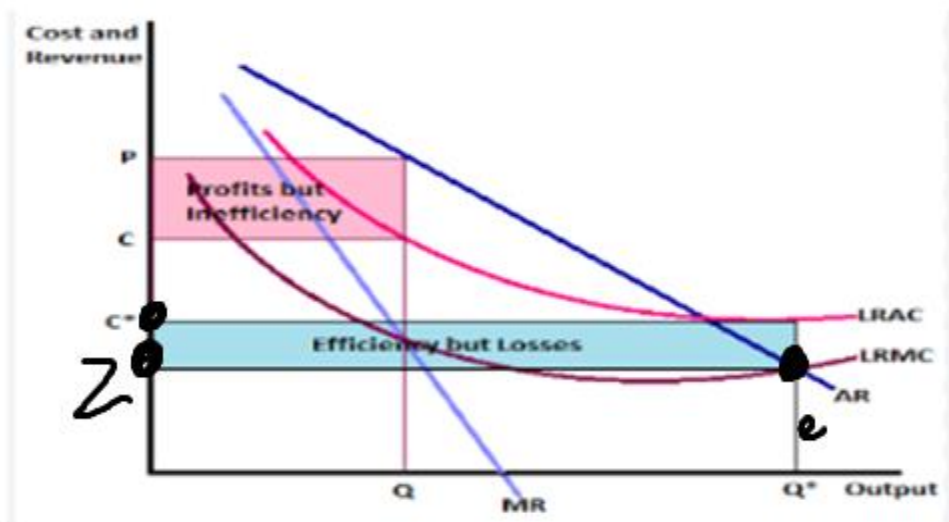
Option C represents the case of “imperfect competition”, whereby there is a downward-sloping demand curve. This can be seen in the diagram below.



This explains why key C is correct.



13 – D:



- A government would set an allocatively efficient level of price i.e. a price of "Z", because that is where, Marginal cost is equal to Average revenue.
- Moreover, here a loss will be made because Average revenue is less than Average costs. This can be seen in the diagram above.
- As a result of this, option D will be correct.

14-D:

A - A free good is a good that is not scarce, and is therefore available without limit for example :sunlight . A free good is available in as great a quantity as desired with zero opportunity cost , therefore it is not charged , as in principle no factors of production are required to produce them, which is why this option is incorrect.

B - Notices do not constitute a direct government regulation. Nudge theory is about guiding behavior through subtle cues rather than imposing regulatory measures. You don't impose or restrict under nudge.

C - Merit goods are goods that generate positive externalities, and their consumption provides benefits to society beyond the individual consumer. The example in the notice is not an example of "merit goods"

D - Nudge theory is a concept from **behavioral economics** that suggests **influencing people's behavior and decisions without mandating or restricting their choices**. The example of the notice on the motorway near Cambridge is an illustration of **nudge theory**.

-> Nudge theory involves **providing information** and by **highlighting that clearing litter puts road-workers' lives at risk**, the notice aims to **increase individuals' awareness** of the **potential negative consequences of their actions** that will encourage them to **make choices that are in their best interest** or align with desired societal outcomes. In this case, the notice about litter being cleared is likely intended to **remind drivers not to litter, thus encouraging a change in behavior to keep the environment clean**.

15 – B:

A – A minimum price for the harmful foods, will result in higher prices, as a result of which, there will be a fall in real income.

B – When a subsidy is provided to substitute foods, as a result of which, they appear cheaper, not only will individuals switch to relatively cheaper substitute foods, which will lead to a fall in consumption of harmful goods, **but also, a fall in prices due to subsidies will result in a rise in real income**. This is why, this option is correct

C and D: An ad-valorem tax(key C) is a type of indirect tax(option D), which results in a fall in supply, and subsequently a rise in prices, as a result of which, there is a fall in real income. **Additionally , indirect taxes are regressive in nature, and they take up a greater proportion of the incomes of the poor, since the rate of tax falls upon a rise in income, which means that they will reduce income equality.**

This explains why key B is correct.

16 – A:

→ **A poverty trap refers to a situation where an individual or household is discouraged from working or earning more income because the financial benefits of doing so are reduced or eliminated by various factors, such as taxes, loss of benefits, or increased expenses.**

Option A describes a situation of gender discrimination in pay, which is unfair but does not create a poverty trap.

The other options describe situations where earning more income leads to reduced net income or increased expenses, making it less likely for the individual to take the higher-paying job:

B) The worker faces a poverty trap due to taxes reducing their net income.

C) The worker faces a poverty trap due to increased expenses (travel and childcare costs) outweighing the higher pay.

D) The worker faces a poverty trap due to the unemployment benefits being more lucrative than the job offer.

These situations illustrate how poverty traps can discourage people from seeking better-paying opportunities, perpetuating cycles of poverty.

17- A:

When labor costs are a small percentage of total cost, the demand for labor in that industry is likely to be inelastic. This means that changes in the wages will not significantly affect the quantity of labor demanded by firms in that industry.

Here's why:

- If labor costs are a small portion of total costs, firms can absorb increases in labor costs without significantly affecting their overall costs or product prices.
- As a result, firms will continue to demand a similar quantity of labor even if wages rise, making the demand for labor inelastic.

B) A price-elastic demand for the final product means that changes in product price will significantly affect quantity demanded, as a result of which, the demand for labor is also likely to be elastic, as firms will not be able to pass their cost burden arising from a rise in wages to consumers in form of a rise in prices.

C) A strong trade union is not likely to result in an inelastic demand for labor.

D) A large pool of readily available labor would result in an increase in labor supply. It will not lead to the demand for labor being inelastic.

18- D:

A) **High saving rates** are likely to promote sustainable economic growth in a developing economy. When people **save more**, it provides a source of capital for **investment in productive activities** in the long-run, such as starting businesses, investing in education and infrastructure, and funding technological advancements.

B) By engaging in international trade, countries can specialize in producing goods and services in which they have a comparative advantage, increasing efficiency and productivity. Free trade can also provide access to new markets, technologies, and investment, which can help drive economic growth and development even in the long-run.

C) Protected property rights are essential for promoting sustainable economic growth in a developing economy. Secure property rights can incentivize landowners and businesses to invest in environmental protection and sustainable practices, as they have a long-term stake in the property and want to maintain its value.

D) Rapid population growth, on the other hand, is least likely to promote sustainable economic growth in a developing economy. A rapidly growing population can put a strain on resources, infrastructure, and services, making it challenging to provide adequate education, healthcare, and employment opportunities. This can lead to poverty, inequality, and social unrest, ultimately hindering economic growth and development

19- B:

A non-material characteristic is the one in which it is difficult to attach a monetary value.

The correct answer is B) the freedom to make informed economic choices.

Non-material characteristics refer to aspects of well-being that go beyond material goods and services, and are difficult to quantify or measure. Examples include:

- ➔ Freedom and autonomy
- ➔ Social connections and relationships
- ➔ Environmental quality
- ➔ Personal fulfillment and happiness

In this case, the freedom to make informed economic choices is an example of a non-material characteristic that affects the standard of living, but is not easily measurable.

The other options are related to material aspects of the standard of living:

A) Unpaid work in the home is an example of non-market activity that is not captured in traditional measures of GDP. This is a material aspect of standard of living.

C) The proportion of GNP used for weapons of war is a material aspect of government spending.

D) The removal of taxes from consumer expenditure is a fiscal policy that affects the cost of living, but is not a non-material characteristic.

20-A:

In this example, the Lorenz curve is moving towards the line of perfect equality, which represents that inequalities have reduced.

This combination of supply-side policies would shift the Lorenz curve from X to Y, indicating a reduction in income inequality:

- Increasing the top rate of income tax reduces the after-tax income of high earners as this represents that the tax system has become more progressive. These funds can be used to provide social security benefits to low-income groups, which in turn helps to redistribute income from higher income groups to lower income groups.
- Free education reduces the barrier to education for low-income individuals, allowing them to acquire skills and increase their earning potential.
- Increasing the national minimum wage raises the floor for low-income workers, ensuring a higher minimum income. This explains why key A will be correct.

21-C:

An economist would classify Mariana's job loss as:

- ➔ Structural unemployment (C): The factory closure due to competition from cheaper imports indicates a structural change in the industry, making Mariana's job skills no longer in demand. This represents international unemployment which is a part of structural unemployment.

And Alfonso's job loss as:

- ➔ Cyclical unemployment (C): The recession-led drop in sales causing Alfonso's job loss is a result of a business cycle downturn, making it a cyclical unemployment case.

22- C: In order to reduce unemployment, we need to employ expansionary fiscal and monetary policy, and also, expansionary supply-side policies.

A – An increase in taxation and interest rates represent contractionary fiscal and monetary policy respectively. Additionally, increasing flexibility in the labor market represents an expansionary supply-side policy.

B – Although increasing government expenditure represents an expansionary fiscal policy and increasing education and training represents an interventionist supply side policy, decreasing money supply is a contractionary monetary policy since it leads to an upward pressure on the interest rate. This explains why key B is incorrect.

C – Decreasing taxation and decreasing interest rate represents expansionary fiscal policy and expansionary monetary policy respectively. An increased flexibility in the labor market is also an expansionary supply side policy. This explains why key C is correct.

D- This is incorrect because decreasing government spending represents a contractionary fiscal policy. However, increasing the money supply represents expansionary monetary policy, and also, increasing education and training represents expansionary supply side policy.

23 – C

A) This combination would have mixed effects on national income. Decreasing direct taxation would increase disposable income, leading to higher consumption, as a result of which, Aggregate demand will rise, which would stimulate economic growth. However, increasing interest rates would make borrowing more expensive, reducing consumption and investment. The net effect would depend on the relative magnitude of the tax cut and interest rate hike.

B) This combination would likely increase national income. Decreasing direct taxation would increase disposable income, leading to higher consumption and Aggregate demand. Additionally, quantitative easing would increase the money supply and since this would ultimately reduce the interest rates, economic activity will be stimulated.

C) This combination would reduce national income. Increasing direct taxation would reduce disposable income, leading to lower consumption and subsequently reduced Aggregate demand. Higher interest rates would make borrowing more expensive, further reducing consumption and investment. This double blow would decrease aggregate demand, leading to lower national income.

D) This combination would have mixed effects on national income. Increasing direct taxation would reduce disposable income, leading to lower consumption and Aggregate demand. However, quantitative easing would increase the money supply, potentially stimulating economic activity after having brought about a decrease in interest rates.

24 – B:

In a **three sector closed economy**, the formula of multiplier will be:

$$\rightarrow 1/(mps+mpt)$$

$$\rightarrow 1/(0.1+0.1) = 5$$

Change in AE (autonomous injection)* multiplier = Change in real national income

$$\rightarrow 100*5 = 500$$

This explains why key B is correct.

The multiplier process explains that increases in autonomous injections in the circular flow will bring about a multiplier times increase in real national income.

25- A:

According to Keynesian liquidity preference theory, the speculative demand for money is influenced by interest rates, as people hold more cash when interest rates are low and hold less cash when interest rates are high as in that case, they prefer to save more in pursuit for higher returns. **Therefore, the combination that is consistent with Liquidity Preference Theory is:**

- Type of demand for money: Speculative
- Determined by: Interest rates
- Determined by changes in income: No

In contrast, transactionary and precautionary demand for money depends on the level of income.

26-B:

B) Keynesians assume that government intervention is necessary to **stabilize the economy**, as they believe that the private sector alone cannot ensure full employment and stable growth. According to Keynesian economics, government spending as part of expansionary fiscal policy can help fill the negative output during economic downturns.

A) Keynesians do not assume that an increase in supply always creates an equal increase in demand. In fact, they argue that aggregate demand may not always be sufficient to match Aggregate supply, as a result of which, demand deficient unemployment is likely to persist even in the long-run.

C) Keynesians do not assume that prices always adjust to clear markets. They recognize that prices may be sticky, especially in the short run, and that markets may not always clear. **In contrast, monetarists believe that prices always adjust so that markets clear.**

D) Keynesians do not assume that the economy always has full employment. In fact, they argue that the economy may experience periods of unemployment, and that government intervention may be necessary to achieve full employment.

In contrast, monetarists believe that the economy will be operating at its potential in the long-run, and it will experience full employment, which means that, there will only be natural rate of unemployment.

27 – B:

Injections represent: government spending, investment spending and export revenues.

- ➔ **No** - If savings are greater than investments, this will lead to a situation whereby leakages are greater than injections, as a result of which, there will be **net leakages from the circular flow.**
- ➔ **No**- If taxes are greater than government spending, this will lead to a situation whereby leakages are greater than injections, as a result of which, there will be **net leakages from the circular flow.**
- ➔ **Yes** – If export revenues are greater than import expenditures, this will lead to a situation whereby injections are greater than leakages, as a result of which, there will be net injections into the circular flow.

28- D:

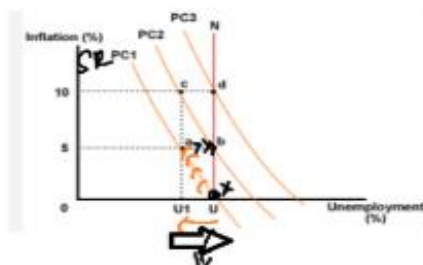
A) This policy would be harmful to developing economies. Resource exploration and extraction can generate significant revenue and create jobs, contributing to economic growth and development. A ban would deprive developing economies of these potential benefits and hinder their economic progress.

B) This policy would also be harmful to developing economies. Foreign investment can bring in capital, technology, and management expertise, helping to develop industries and create employment opportunities. Restrictions would limit access to these benefits, hindering economic development.

C) This policy could also lead to a brain drain, depriving developing economies of the skills and expertise they need to develop their own industries and economies.

D) This policy would be beneficial for developing economies. Agricultural subsidies in developed economies can create an uneven playing field, making it difficult for farmers in developing economies to compete. Withdrawal of these subsidies would create opportunities for farmers in developing economies to export their products and grow their agricultural sectors. This explains why key D will be correct.

29- A:



* $2\downarrow \rightarrow S\downarrow - C\uparrow - AD\uparrow \rightarrow$ movement along $SRPC_1$ from X to $a \rightarrow U_n\downarrow (U-U_1) \rightarrow$ inflation rises (0% to 5%)
 * $AD\uparrow$, production $\uparrow \rightarrow D\uparrow$, wages $\uparrow$
 * wages $\uparrow >$ productivity $\rightarrow$ costs of production rise, $AS\downarrow$, philips curve shifts rightwards from $SRPC_1$ to $SRPC_2$ (movement from a to b).
 * In the long-run, the economy reaches to the $NAIRU$ again, if demand side policies are (U) used.

1. **Government tries to reduce unemployment below NAIRU:** The government uses expansionary monetary policy to stimulate the economy and reduce unemployment below the natural rate (NAIRU).
2. **Reduction in unemployment:** Unemployment falls as the economy grows and more people find jobs.
3. **Upward movement along SRPC (higher inflation):** As unemployment falls, the economy moves upward along the Short-Run Phillips Curve (SRPC), and there is a rise in demand pull inflation.

Higher inflation results from the increased demand for labor and resources. This shows the trade-off between unemployment and inflation.

4. **Short-run Philips curve shifts towards the right:** When there is a mismatch between wage increases and increases in productivity, there will be an increase in costs of production, as a result of which, Aggregate supply will fall, leading to a rightward shift in the short run Philips curve. This indicates an increase in the rate of inflation and also unemployment as the economy moves back to the natural rate of unemployment.
5. **Long-run adjustment: economy returns to NAIRU, inflation stabilizes at a higher level:** Eventually, the economy adjusts to the demand-side shocks, and the labor market returns to equilibrium at the natural rate of unemployment.

This is why only key A (**accelerating rate of inflation**) will be correct.

30 – A:

-Reflationary policies represent expansionary policies.

-Deflationary policies represent contractionary policies.

The correct answer is A) increasing government spending.

A reflationary fiscal measure is a policy aimed at stimulating economic activity and countering deflation. Increasing government spending is a classic example of a reflationary fiscal measure, as it boosts aggregate demand, stimulates economic growth, and helps combat deflationary pressures.

The other options are not correct:

B) Increasing taxes represent **contractionary fiscal policy** that would have a deflationary effect, reducing aggregate demand and economic activity.

C) Increasing the money supply is a reflationary monetary policy tool, not a fiscal measure.

D) Reducing interest rates is also a reflationary monetary policy tool, not a fiscal measure.

